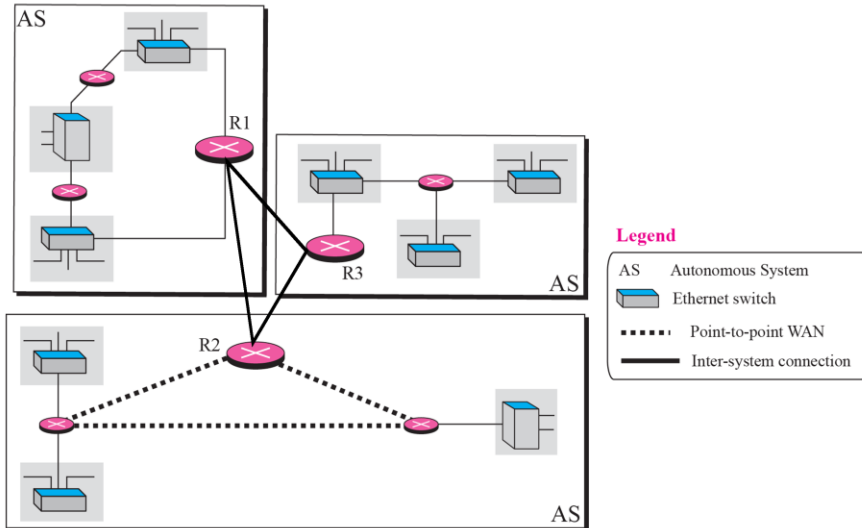
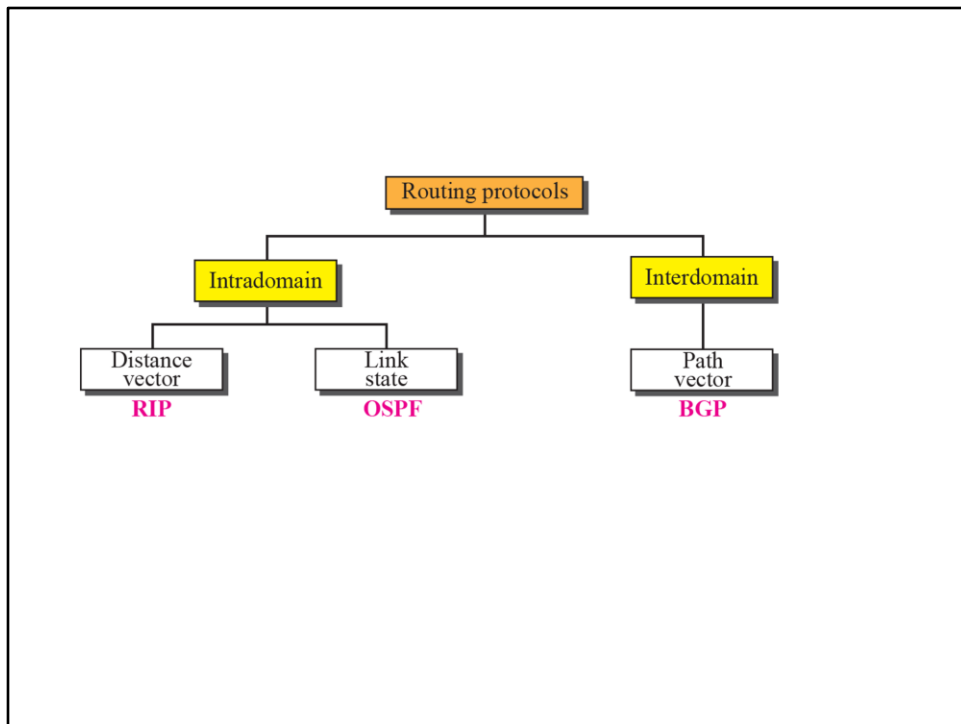


Unicast Routing Protocols

Autonomous systems



- An autonomous system (AS) is a group of networks and routers under the authority of a single administration.
- Routing inside an autonomous system is called **intra-domain routing**.
- Routing between autonomous systems is called **inter-domain routing**

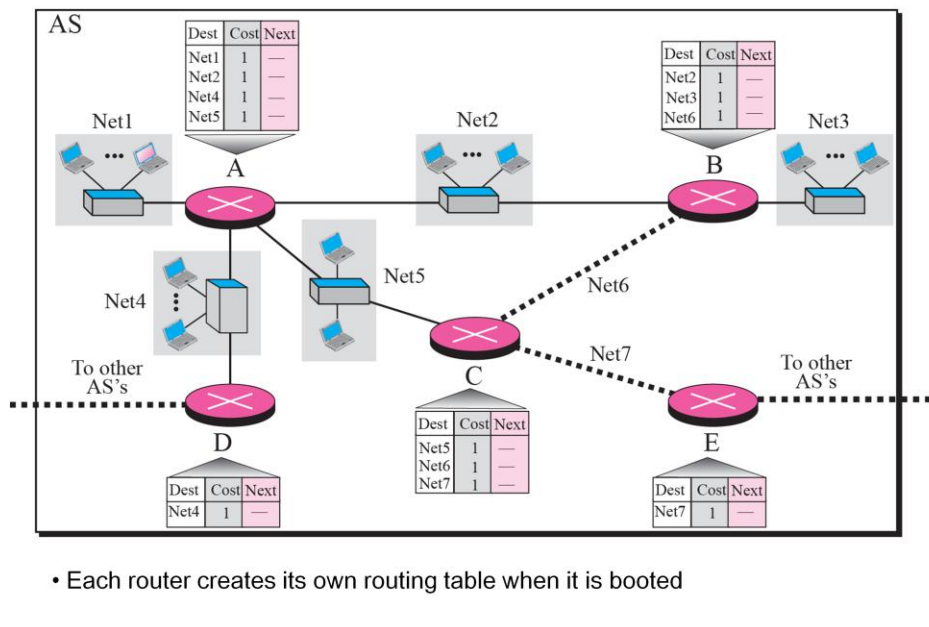


DISTANCE VECTOR ROUTING

- Initialization
- Sharing
- Updating

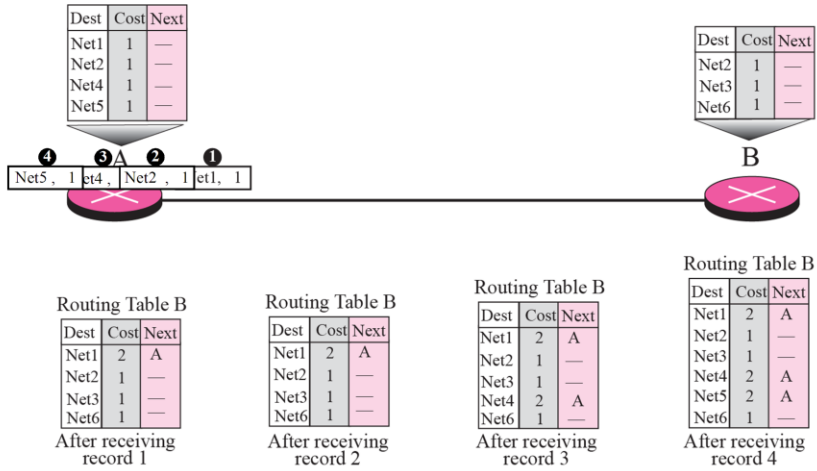
In distance vector routing, each node shares its routing table with its immediate neighbors periodically and when there is a change.

Example



Now assume router A sends four records to its neighbors, routers B, D, and C.

Below figure shows the changes in B's routing table when it receives these records.



Final routing tables for routers

A

Dest	Cost	Next
Net1	1	—
Net2	1	—
Net3	2	B
Net4	1	—
Net5	1	—
Net6	2	C
Net7	2	C

B

Dest	Cost	Next
Net1	2	A
Net2	1	—
Net3	1	—
Net4	2	A
Net5	2	A
Net6	1	—
Net7	2	C

C

Dest	Cost	Next
Net1	2	A
Net2	2	A
Net3	2	B
Net4	2	A
Net5	1	—
Net6	1	—
Net7	1	—

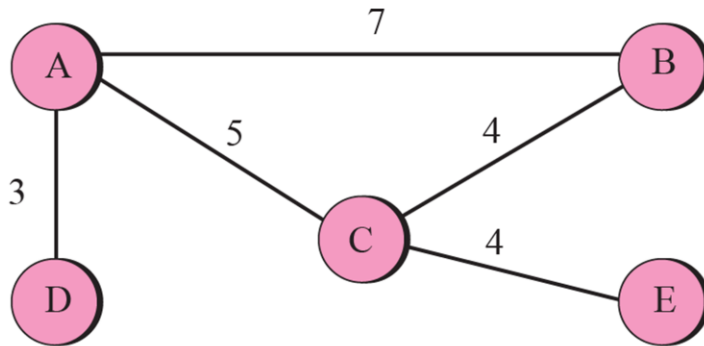
D

Dest	Cost	Next
Net1	2	A
Net2	2	A
Net3	3	A
Net4	1	—
Net5	1	A
Net6	3	A
Net7	3	A

E

Dest	Cost	Next
Net1	3	C
Net2	3	C
Net3	3	C
Net4	3	C
Net5	2	C
Net6	2	C
Net7	1	—

Construct the routing table of each router in the below figure



Routing Information Protocol (RIP)

- intra-domain (interior) routing protocol used inside an autonomous system.
- based on distance vector routing.

RIP messages

Request

- A request message is sent by a router that has just come up or by a router that has some time-out entries
- A request can ask about specific entries or all entries

Response

- A response can be either solicited or unsolicited (30s or when there is a change in the routing table)

RIP Timers

Periodic timer

- It controls the advertising of regular update message (25 ~ 30 sec)

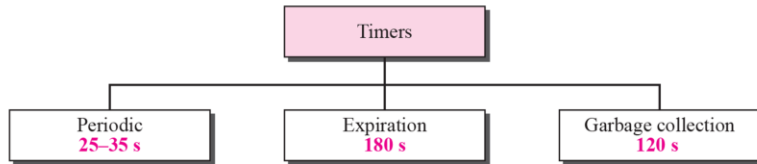
Expiration timer

- It governs the validity of a route (180 sec).
- The route is considered expired and the hop count of the route is set to 16

Garbage collection timer

- An invalid route is not purged from the routing table until this timer expires (120 sec)

RIP timers



Example

A routing table has 20 entries. It does not receive information about five routes for 200 s. How many timers are running at this time?

Solution

The 21 timers are listed below:

Periodic timer: 1

Expiration timer: $20 - 5 = 15$

Garbage collection timer: 5